Endpoint-collocation spectral amplitudes for scalar Teukolsky scattering on Kerr

KerrScattering Project[1]

[1] *Manuscript draft generated from the KerrScattering repository*

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We present a direct Chebyshev spectral construction of scalar ($s = 0$) Teukolsky scattering amplitudes on a Kerr black hole. The method is designed as the Kerr continuation of a Schwarzschild Green-function spectral code: the compact coordinate is $z = r_+/r$, the physical endpoints $z = 0$ and $z = 1$ are retained as collocation nodes, and the degeneracy of the second-order radial equation at both endpoints is used as a regular first-order boundary condition. With the horizon normalization $B^{\rm trans} = 1$, the incident and reflected amplitudes are obtained by matching a horizon-normalized in solution to down/up solutions at an intermediate radius. The invariant validation quantities are $|B^{\rm inc}|$, $|B^{\rm ref}|$, and the Kerr flux balance relation; complex phases are convention dependent because of the additive constant in the Kerr tortoise coordinate. Across 37 independent GeneralizedSasakiNakamura.jl benchmark cases, including nonrotating, moderate-spin, near-superradiant, and high-spin configurations, the worst amplitude-magnitude discrepancy is 1.485×10^{-8} and the worst flux residual is 3.714×10^{-9}. A 51-point frequency sweep resolves scalar superradiance, with amplification appearing only for $\omega < m\Omega_H$. We also describe the implemented spheroidal cubic projector and a first Green-function radial diagnostic for weak $|\Phi|^2\Phi$ sources. The latter is a diagnostic framework rather than a final nonlinear Kerr normalization, and it identifies the remaining task: deriving the covariant Kerr radial source weight and replacing finite diagnostic quadrature by controlled oscillatory-tail integration.

I. INTRODUCTION

The Teukolsky formalism separates massless field perturbations on a Kerr black hole into angular spheroidal harmonics and a radial ordinary differential equation [1]. For scalar fields the system is already rich enough to test the central numerical issues that appear in Kerr scattering: horizon phase normalization, superradiant flux balance, spheroidal angular mixing, and the conditioning of small reflected amplitudes. The purpose of this work is not to introduce a new radial equation, but to make the numerical continuation from a Schwarzschild spectral Green-function code to Kerr precise and reproducible.

The central design choice is to keep the physical boundaries in the collocation grid. In the compact coordinate $z = r_+/r$, infinity is at $z = 0$ and the event horizon is at $z = 1$. After peeling off the known oscillatory asymptotic factor, the radial equation becomes singularly regular: the coefficient of the second derivative vanishes at each endpoint, leaving a first-order condition. This mirrors the endpoint treatment in the Schwarzschild spectral method and avoids artificial horizon or infinity truncation.

The implementation is validated against GeneralizedSasakiNakamura.jl in the unit-Teukolsky-transmission convention. We use only invariant quantities for the pass/fail comparison: amplitude magnitudes and the flux balance residual. The complex amplitudes themselves are not invariant under a shift $r_* \to r_* + C$ and therefore differ by constant phases across packages.

The remainder of this paper is organized as follows. Section II sets the scalar Teukolsky conventions. Section III describes the endpoint spectral discretization and amplitude extraction. Section IV gives the benchmark and frequency-sweep results. Section V describes the cubic spheroidal projector and the current nonlinear Green-function diagnostic. Section VI summarizes the separate $s = -2$ benchmark side task. We conclude in Sec. VII.

II. SCALAR TEUKOLSKY EQUATION ON KERR

A. Geometry and tortoise coordinate

In Boyer–Lindquist coordinates,

$$\Sigma = r^2 + a^2 \cos^2\theta, \tag{1}$$

$$\Delta = r^2 - 2Mr + a^2 = (r - r_+)(r - r_-), \tag{2}$$

$$r_+ = M + \sqrt{M^2 - a^2}, \qquad r_- = M - \sqrt{M^2 - a^2}. \tag{3}$$

The horizon angular velocity is

$$\Omega_H = \frac{a}{r_+^2 + a^2} = \frac{a}{2Mr_+}. \tag{4}$$

We use the Kerr tortoise coordinate

$$\frac{\mathrm{d}r_*}{\mathrm{d}r} = \frac{r^2 + a^2}{\Delta}, \tag{5}$$

so that

$$r_* = r + \frac{2Mr_+}{r_+ - r_-}\ln(r - r_+) - \frac{2Mr_-}{r_+ - r_-}\ln(r - r_-) + C. \tag{6}$$

The constant C fixes a phase convention but has no invariant content.

B. Angular equation and separation constant

For a massless complex scalar field, $\Box\Phi = 0$, we separate

$$\Phi(t, r, \theta, \phi) = R_{\ell m\omega}(r) S_{\ell m}(\theta; a\omega) e^{im\phi - \mathrm{i}\omega t}. \tag{7}$$

The scalar spheroidal function satisfies

$$0 = \frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}S_{\ell m}}{\mathrm{d}\theta}\right) + \left[a^2\omega^2\cos^2\theta - \frac{m^2}{\sin^2\theta} + A_{\ell m}\right] S_{\ell m}. \tag{8}$$

For $c = a\omega \to 0$, $A_{\ell m} \to \ell(\ell + 1)$ and $S_{\ell m}$ reduces to the corresponding spherical harmonic with the $e^{im\phi}$ factor removed. The radial separation constant used here is

$$\lambda = A_{\ell m} + a^2\omega^2 - 2am\omega. \tag{9}$$

This convention is recorded explicitly in the output files, because different packages return either $A_{\ell m}$ or a radial separation constant.

C. Radial equation and asymptotics

Defining

$$K(r) = (r^2 + a^2)\omega - am, \tag{10}$$

the scalar radial Teukolsky equation is

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(\Delta\frac{\mathrm{d}R}{\mathrm{d}r}\right) + \left(\frac{K^2}{\Delta} - \lambda\right) R = 0. \tag{11}$$

Near the horizon the physical frequency is

$$p = \omega - m\Omega_H. \tag{12}$$

The in solution is normalized by

$$R_{\mathrm{in}} \sim B^{\mathrm{trans}} e^{-\mathrm{i}pr_*}, \qquad B^{\mathrm{trans}} = 1, \qquad r \to r_+. \tag{13}$$

At infinity,

$$R_{\mathrm{in}} \sim B^{\mathrm{inc}}\frac{e^{-\mathrm{i}\omega r_*}}{r} + B^{\mathrm{ref}}\frac{e^{+\mathrm{i}\omega r_*}}{r}. \tag{14}$$

The up solution is purely outgoing at infinity,

$$R_{\mathrm{up}} \sim \frac{e^{+\mathrm{i}\omega r_*}}{r}. \tag{15}$$

D. Wronskian and flux

Equation (11) has a conserved radial Wronskian

$$W[R_1, R_2] = \Delta\left(R_1\frac{\mathrm{d}R_2}{\mathrm{d}r} - R_2\frac{\mathrm{d}R_1}{\mathrm{d}r}\right). \tag{16}$$

For the normalization (13), the scalar flux relation is

$$\mathcal{R} + \mathcal{T} = 1, \tag{17}$$

with

$$\mathcal{R} = \frac{|B^{\mathrm{ref}}|^2}{|B^{\mathrm{inc}}|^2}, \qquad \mathcal{T} = \frac{p(r_+^2 + a^2)}{\omega}\frac{|B^{\mathrm{trans}}|^2}{|B^{\mathrm{inc}}|^2}. \tag{18}$$

For $\omega < m\Omega_H$, $\mathcal{T} < 0$ and therefore $\mathcal{R} > 1$, which is the scalar superradiant regime [2].

III. ENDPOINT CHEBYSHEV SPECTRAL CONSTRUCTION

A. Compactification and phase peeling

We use

$$z = \frac{r_+}{r}, \qquad z = 0 \text{ at } \mathcal{I}^+, \qquad z = 1 \text{ at } H^+. \tag{19}$$

Each homogeneous branch is written as

$$R(z) = F_{\mathrm{branch}}(z) u(z), \tag{20}$$

where F_{branch} contains the known asymptotic phase:

$$F_{\mathrm{down}} = e^{-\mathrm{i}\omega r_*}/r, \tag{21}$$

$$F_{\mathrm{up}} = e^{+\mathrm{i}\omega r_*}/r, \tag{22}$$

$$F_{\mathrm{in}} = e^{-\mathrm{i}pr_*}. \tag{23}$$

The reduced function $u(z)$ is smooth at the relevant endpoint and is normalized by $u = 1$ there.

After substitution, each branch has the schematic equation

$$B_2(z)u'' + B_1(z)u' + B_0(z)u = 0. \tag{24}$$

At $z = 0$ and $z = 1$, B_2 vanishes. The collocation matrix therefore keeps the degenerate endpoint row as a first-order regularity condition and uses the neighboring row to impose $u(\text{endpoint}) = 1$. No finite-radius horizon or infinity cutoff is introduced.

B. Two-domain matching

The implementation uses two Chebyshev domains, $[0, z_m]$ and $[z_m, 1]$. On the outer domain we solve the down and up branches. On the inner domain we solve the horizon-normalized in branch. At $z_m = r_+/r_m$ the physical field and radial derivative satisfy

$$R_{\mathrm{in}} = B^{\mathrm{inc}} R_{\mathrm{down}} + B^{\mathrm{ref}} R_{\mathrm{up}}, \tag{25}$$

$$\frac{\mathrm{d}R_{\mathrm{in}}}{\mathrm{d}r} = B^{\mathrm{inc}}\frac{\mathrm{d}R_{\mathrm{down}}}{\mathrm{d}r} + B^{\mathrm{ref}}\frac{\mathrm{d}R_{\mathrm{up}}}{\mathrm{d}r}. \tag{26}$$

Thus B^{inc} and B^{ref} are obtained from a 2×2 linear system. The matching radius and spectral order are selected adaptively in the validation scripts, with sinh clustering used for low-frequency cases when useful.

TABLE I. Validation summary for the scalar Kerr spectral solver. The "score" is the maximum of the invariant amplitude-magnitude errors and the flux residual.

Quantity	Value
GSN benchmark requests	37
Usable GSN cases	37
Excluded GSN cases	0
All GSN cases passed	true
Worst score	1.485×10^{-8}
Worst flux residual	3.714×10^{-9}
Frequency-sweep points	51
Worst sweep score	9.467×10^{-8}
Worst sweep flux residual	7.351×10^{-9}
Maximum sampled $\mathcal{R} - 1$	4.970×10^{-4}

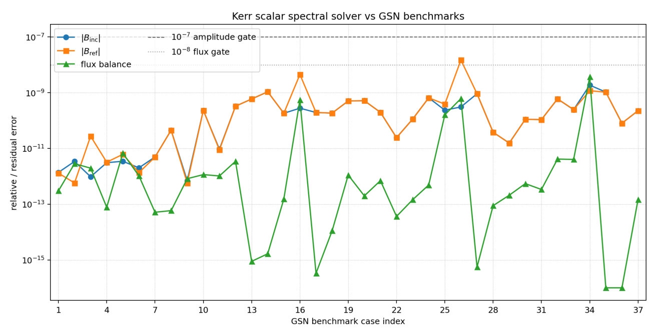

FIG. 1. Invariant GSN benchmark errors for the validated scalar Kerr grid. Only amplitude magnitudes and flux balance are used for pass/fail decisions.

IV. LINEAR VALIDATION

A. GSN benchmark protocol

The external benchmark is GeneralizedSasakiNakamura.jl [5] in the `UNIT_TEUKOLSKY_TRANS` convention. We compare $|B^{\rm inc}|$, $|B^{\rm ref}|$, and the flux residual. The complex phases are retained in CSV output but not used as benchmark errors, because a shift in r_* maps

$$B^{\rm inc} \to B^{\rm inc} e^{-{\rm i}\omega C}, \qquad B^{\rm ref} \to B^{\rm ref} e^{+{\rm i}\omega C}. \tag{27}$$

The acceptance gates are

$$\frac{\delta|B^{\rm inc}|}{|B^{\rm inc}|} \le 10^{-7}, \qquad \frac{\delta|B^{\rm ref}|}{|B^{\rm ref}|} \le 10^{-7},$$
$$|\mathcal{R} + \mathcal{T} - 1| \le 10^{-8}. \tag{28}$$

The benchmark grid covers the Schwarzschild limit, moderate Kerr rotation, near-threshold scalar superradiance, low-frequency superradiant scattering, high-frequency small-reflection scattering, and high-spin cases. The hardest case is $(a, \ell, m, \omega) = (0.5, 2, 2, 1.0)$, where the reflected amplitude is small and the relative error is correspondingly sensitive to conditioning.

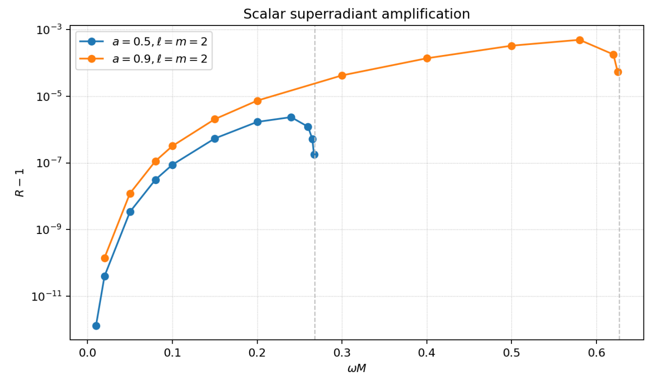

FIG. 2. Sampled scalar superradiant amplification. Positive amplification is confined to the region $\omega < m\Omega_H$.

B. Frequency sweep and superradiance

The frequency sweep contains one Schwarzschild reference sequence and two Kerr sequences:

$$a = 0, \quad \ell = m = 0, \quad 11 \text{ points}, \tag{29}$$
$$a = 0.5, \quad \ell = m = 2, \quad 20 \text{ points}, \tag{30}$$
$$a = 0.9, \quad \ell = m = 2, \quad 20 \text{ points}. \tag{31}$$

For the rotating cases the sampled amplification $\mathcal{R} - 1$ is positive only for $\omega < m\Omega_H$, in agreement with Eq. (18).

V. CUBIC ANGULAR PROJECTOR AND NONLINEAR DIAGNOSTIC

For a weak complex scalar self-interaction,

$$\Box\Phi + \epsilon|\Phi|^2\Phi = 0, \tag{32}$$

a single incident mode sources the angular structure $|S_{\ell m}|^2 S_{\ell m}$. This must be projected onto the spheroidal basis:

$$|S_{\ell m}|^2 S_{\ell m} = \sum_{\ell'} C_{\ell' \ell m} S_{\ell' m}, \tag{33}$$

where

$$C_{\ell' \ell m} = \int {\rm d}\Omega\, S^*_{\ell' m} |S_{\ell m}|^2 S_{\ell m}. \tag{34}$$

The code evaluates Eq. (34) for every validated source mode and target channels $|m| \le \ell' \le \ell + 4$. The present table contains 199 coupling rows from 37 source modes. The self-channel coefficient ranges from 7.958×10^{-2} to 1.705×10^{-1}, while the largest off-diagonal coefficient is 8.318×10^{-2} for $(\ell, m, a, \omega, \ell') = (2, 0, 0.5, 0.1, 4)$. The spherical checks $C_{000} = 1/(4\pi)$ and $C_{110} = 9/(20\pi)$ are satisfied.

The nonlinear radial diagnostic uses the validated homogeneous solutions and the conserved Wronskian to

TABLE II. Prototype nonlinear Green-function diagnostic amplitudes at quadrature order $q = 192$. The amplitudes use the diagnostic `legacy-bondi-dr-over-r2` radial weight and should not yet be read as final Kerr nonlinear scattering observables. Full complex values are stored in `results/kerr_scalar_nonlinear_diagnostics.csv`.

case	(a, ω)	$\lvert A^{(1)}_{\rm ref}\rvert$	$\lvert A^{(1)}_{H}\rvert$	$\max \delta A$
Schw. $(0,0)$	$(0, 0.10)$	2.808×10^{-2}	1.704×10^{-2}	2.103×10^{-5}
Kerr sub.	$(0.5, 0.24)$	7.780×10^{2}	1.836	5.083×10^{-7}
Kerr super.	$(0.5, 0.30)$	1.197×10^{2}	7.645×10^{-1}	9.627×10^{-7}
High spin	$(0.9, 0.58)$	1.298×10^{1}	5.683×10^{-1}	3.164×10^{-6}

compute prototype Green-function source integrals. At present the radial source weight is explicitly labelled `legacy-bondi-dr-over-r2`; it is a diagnostic bridge from the Schwarzschild code, not a final Kerr self-interaction convention. With four representative cases and quadrature orders $64, 96, 128, 192$, the worst Wronskian consistency error is 3.094×10^{-16}. The worst consecutive quadrature changes are 7.011×10^{-5} for the reflected correction and 8.331×10^{-5} for the horizon correction. The run is resource guarded; the recorded peak process RSS for the current diagnostic table is 83.0 MB.

These diagnostics show that the homogeneous Green-function skeleton is stable. The remaining nonlinear physics is the derivation of the Kerr covariant radial source weight and the replacement of finite diagnostic quadrature by an oscillatory-tail method with explicit convergence gates.

VI. SPIN-MINUS-TWO BENCHMARK SIDE TASK

The repository also records a separate $s = -2$ benchmark task for $a = 0.5$, $\ell = m = 2$, and $\omega = 10^{-4}, 0.1, 10$. Low and intermediate frequencies are compared against a Mathematica MST route [4], while the high frequency case uses GeneralizedSasakiNakamura.jl. The direct Teukolsky-variable Python comparison gives relative errors

$$\omega = 10^{-4}: \quad \delta B^{\rm inc}/B^{\rm inc} = 2.5 \times 10^{-8}, \quad \delta B^{\rm ref}/B^{\rm ref} = 9.1 \times 10^{-12}, \tag{35}$$

$$\omega = 0.1: \quad \delta B^{\rm inc}/B^{\rm inc} = 2.7 \times 10^{-10}, \quad \delta B^{\rm ref}/B^{\rm ref} = 5.3 \times 10^{-12}, \tag{36}$$

$$\omega = 10: \quad \delta B^{\rm inc}/B^{\rm inc} = 4.0 \times 10^{-7}, \quad \delta B^{\rm ref}/B^{\rm ref} = 3.4 \times 10^{-1}. \tag{37}$$

The high-frequency reflected amplitude is approximately 10^{-8}, close to the double-precision floor for direct Teukolsky-variable matching. Pushing that case to benchmark accuracy should be done in the Sasaki–Nakamura variable [3], followed by the Teukolsky conversion. This side task is therefore kept separate from the scalar solver.

VII. CONCLUSIONS

We have constructed and validated a Kerr scalar scattering solver that follows the same endpoint-collocation philosophy as the Schwarzschild spectral Green-function code. The key points are: the compact coordinate $z = r_+/r$ keeps both physical boundaries in the grid; phase peeling converts the endpoint singularities into regular first-order rows; horizon-normalized in modes are matched to down/up modes to extract $B^{\rm inc}$ and $B^{\rm ref}$; and the invariant flux balance provides a stringent convention-independent check.

The linear scalar problem is closed at the level required for a publication draft: 37 GSN benchmark cases and 51 frequency-sweep points pass the stated acceptance gates, and the sweep resolves the expected Kerr superradiant amplification. The cubic angular projector and first radial Green-function diagnostics are implemented, but the nonlinear source normalization is not yet final. The next physics step is to derive the Kerr radial source weight from the covariant scalar equation and to introduce a controlled oscillatory-tail integration method for nonlinear amplitudes.

ACKNOWLEDGMENTS

This draft was generated from the numerical outputs in the KerrScattering repository on the branch `codex/teukolsky-scalar-derivation`. The external scalar benchmark uses GeneralizedSasakiNakamura.jl.

DATA AND CODE AVAILABILITY

The validation data are stored in CSV files under `results/`. The main files used in this manuscript are `kerr_scalar_adaptive_summary.csv`, `kerr_scalar_frequency_sweep.csv`, `kerr_scalar_cubic_couplings.csv`, and `kerr_scalar_nonlinear_diagnostics.csv`.

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